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A series of recommended tests when validating probabilistic DNA profile interpretation software

Jo-Anne Bright [a,b,*], Ian W. Evett [c], Duncan Taylor [d,e], James M. Curran [b], John Buckleton [a]

[a] ESR, Private Bag 92021, Auckland 1142, New Zealand
[b] University of Auckland, Department of Statistics, Private Bag 92019, Auckland 1142, New Zealand
[c] Principal Forensic Services Ltd, United Kingdom
[d] Forensic Science South Australia, 21 Divett Place, SA 5000, Australia
[e] School of Biological Sciences, Flinders University, GPO Box 2100, Adelaide, SA 5001, Australia

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ABSTRACT

There has been a recent push from many jurisdictions for the standardisation of forensic DNA interpretation methods. Current research is moving from threshold-based interpretation strategies towards continuous interpretation strategies. However laboratory uptake of software employing probabilistic models is slow. Some of this reluctance could be due to the perceived intimidating calculations to replicate the software answers and the lack of formal internal validation requirements for interpretation software. In this paper we describe a set of experiments which may be used to internally validate in part probabilistic interpretation software. These experiments included both single source and mixed profiles calculated with and without dropout and drop-in and studies to determine the reproducibility of the software with replicate analyses. We do this by way of example using three software packages: STRmix™, LRmix, and Lab Retriever. We outline and demonstrate the profile examples where the expected answer may be calculated and provide all calculations.

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1. Introduction

The amplification of short tandem repeats (STRs) by the polymerase chain reaction (PCR) is the dominant method of forensic DNA analysis. DNA evidence interpretation is widely accepted in criminal investigations, however there are still recent examples where existing practices have been questioned [1,2]. Prevalent DNA interpretation strategies have been shown to result in likelihood ratios that vary between different laboratories by orders of magnitude [3,4].

There is pressure from many jurisdictions for standardisation including within Australasia, the US [5], and Europe [6]. The International Society for Forensic Genetics (ISFG) have published recommendations for the interpretation of mixed DNA profiles [7], Y STR profiles [8] and profiles that may have allelic dropout or drop-in [9]. In 2012, the editors of Forensic Science International: Genetics (affiliated with ISFG) encouraged more research and the creation of statistical software packages to advance the development and implementation of generally accepted standards for forensic genetics [10].

Current research is steering away from threshold-based interpretation strategies and towards continuous interpretation strategies [11–14]. The introduction of semi and fully continuous (probabilistic) methods has the potential to increase the efficiency of forensic laboratories, and improve the consistency and transparency of the reported results. These probabilistic programmes calculate a likelihood ratio (*LR*). The *LR* is widely considered to be the most powerful and relevant measure of the weight of evidence [15]. It is the ratio of the probability of the evidence (E) given each of two competing hypotheses, H_p, which typically aligns with the prosecution hypotheses, and H_d, which aligns with the defence hypothesis:

$$LR = \frac{Pr(E|H_p)}{Pr(E|H_d)}$$

Semi-continuous methods (also known as discrete or drop models) can optionally incorporate a probability for dropout ($Pr(D|R)$) and/or a probability for drop-in ($Pr(C|S)$), where R is the information used to assign the probability of dropout and S the information used to assign the probability of drop-in. We will

* Corresponding author at: ESR, Private Bag 92021, Auckland 1142, New Zealand. Tel.: +64 8153940; fax: +64 98496046.
E-mail address: Jo.bright@esr.cri.nz (J.-A. Bright).

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sometimes omit the conditioning information in future use of these terms. These semi-continuous methods do not use peak heights when generating possible genotype sets and do not model artefacts such as stutter. In these models, peaks must be assigned as stutter or labelled as allelic by an analyst prior to interpretation. One programme [13] allows an ambiguous peak designation of either stutter or allele which seems to be a real advantage. Lab Retriever implements this approach by characterizing the ambiguous peak as a masking allele, and we have found that the same approach may be applied to LRmix (see Appendix 3 for a more complete description of this implementation). The probability of the evidence given all possible genotype sets is then calculated [15,16].

Fully continuous methods assign a probability density for the observed profile or profiles (if interpreting multiple amplifications or replicates) given all the possible genotype combinations. A continuous approach uses nearly all the information within a profile, including peak height, and models the uncertainty in the behaviour of peaks including back stutter. Continuous systems therefore do not require the analyst to specify whether they believe a peak in a back stutter position is allelic or stutter (or a composite of the two) as these possibilities will be considered in the analysis. Both semi and fully continuous based methods are software-based solutions because of their complexity. The replication of the complex calculations by hand is potentially daunting for laboratories and this factor is plausibly preventing uptake of the models.

The requirements for the internal validation of interpretation software have not been previously outlined. We make a distinction between the requirements for developmental versus internal validation. This distinction is also made in the Quality Assurance Standards for Forensic DNA Testing Laboratories (QAS) [17]. Developmental validation is described as (in part) the determination of conditions and limitations of a new or novel DNA methodology for use on forensic samples. It does not explicitly reference interpretation software. Internal validation is data generated within a laboratory to demonstrate that established methods perform as expected. The QAS standards state that interpretation guidelines, including guidelines for mixture interpretation, shall be established using the findings of the internal validation. The Scientific Working Group on DNA Analysis Methods Validation Guidelines (2012) contain recommendations for the validation of DNA analysis methods including software [18], however no guidance is provided on what tests should be undertaken. Internal validation studies traditionally involve tests of accuracy and reproducibility.

There is a subset of profiles where the expected *LR* may be replicated relatively easily by hand. We use the word "expected" here with care. The expected answer from a software programme is the one that should be produced using the model in use. It would be wrong to substitute the words "true" or "correct" as the answer is one produced by a model and the reliability of the model is a matter of subjective judgement.

The performance and limitations of the software may be examined by comparing the software output with the expected answer. An understanding of the models behind the methods is essential for this process as is an understanding of the limitations of the different methods. Examples of where we can predict the expected answer include single source profiles, mixtures where the profile of a major contributor is unambiguous (major/minor) and mixtures of two contributors in equal proportions (balanced). In a simplification of the calculations, profiles may be analysed where no dropout or drop-in is assumed. In subsequent trials, the effect of adding dropout and drop-in on the *LR* can be determined. A performance requirement for any software product should be that its output is intuitively sensible to the analyst given the inputs. A large *LR* does not, in itself, satisfy this requirement.

In this paper we recommend a set of experiments that may be used in part to internally validate software. We suggest they are also essential during developmental validation. We do this by way of example using three software packages: STRmix™ [11], LRmix [19], and Lab Retriever [20]. We outline and demonstrate the profile examples where the expected answer may be calculated. We propose that these validation profiles should be artificially generated by means of a mathematical model rather than from profiling of made up samples. This approach has previously been advocated by Gill et al. [21]. In this way we can control the input variables exactly and produce profiles where the expected answer is known. Consider that it is effectively impossible to create an exact 1:1 mixture in vitro. These experiments would not investigate the full range of functionality of most probabilistic systems but test whether the allele probabilities, population genetic model, and genotype combinations are performing as expected. The mathematics used to construct the profiles in this paper are not the same model as used in any of the programmes except in as much that stutter models used to create the profiles and some decision rules for handling stutter and assigning probabilities for the application of all three programmes have evolved from the same empirical basis. We assert that there is no preselection bias in the creation of the profiles that assists or restricts one or more of these programmes. They are simply tidy single source, major:minor and balanced profiles. Recall the goal of this paper is to specify tests that could and, we argue, should be applied to any software. This is not a head-to-head test of software. In order to "know" the expected answer the profiles need to be simplified. Real samples seldom display this simplicity.

Lab Retriever and LRmix are semi-continuous models. They both require the user to enter a value for the probability of dropout and drop-in. They do not use peak heights and do not model stutter peak heights. LRmix and Lab Retriever input files do not require pre-processing of stutters peaks, but if not filtered, they usually lead to an underestimation of the *LR*, since the sample allele count increases. The peaks are alternatively treated as true and/or drop-in alleles.

STRmix™ uses a fully continuous model for DNA profile interpretation [11] modelling the mass of an allele by fitting an exponential degradation curve to each contributor within a profile [22]. A 'per allele' stutter ratio is applied [23,24]. STRmix™ assigns a weight to each possible genotype combination at a locus. The weights across all combinations at that locus sum to one. A single unambiguous genotype combination at any locus would therefore be assigned a weight of one. The weights are subsequently incorporated into the likelihood ratio calculation. STRmix™ and LRmix implement the same population genetic model namely the Balding and Nichols' equations also known as recommendation 4.2 of NRC II [25,26]. Lab Retriever implements approximations present in the Balding and Buckleton paper [13].

The desirability of using one interpretation model that can be applied across all template amounts has been emphasised repeatedly in the recent literature [6,27–29]. It is preferable that this model can be applied without restrictive workarounds such as locus dropping. All three software products examined here meet this condition. Maintaining two or more methods imposes a quality assurance and training load on an organisation that is unnecessary and counterproductive.

2. Method

2.1. Software

STRmix™ V2.0 is commercial software (www.strmix.esr.cri.nz). Lab Retriever and LRmix are both free open source software available on the internet. Version 1.2.4 of the Lab Retriever

software was downloaded from the Scientific Collaboration, Innovation and Education (SCIEG) website (http://www.scieg.org/lab_retriever.html). LRmix is a module from within the R package forensim. Version 4.3 of forensim and its dependencies were downloaded from the Comprehensive R Archive Network (CRAN) website http://www.r-project.org.

2.2. Profile generation

Mixed DNA profiles were created artificially based on two single source Applied Biosystems' Identifiler™ profiles analysed on an Applied Biosystems' 3130 capillary electrophoresis instrument. They were created mathematically by combining the two single source profiles in specific ratios and choosing template levels by creating a set of artificial peak heights. Stutter was added to the profile at a per allele rate [24]. Allele repeat number, height in relative fluorescent units (rfu) and size were retained in the STRmix™ input file as per the user's manual. Height and size information was removed from the LRmix and Lab Retriever input files. Prior to interpretation of the single source profiles, peaks in stutter positions with heights less than the Applied Biosystems' per locus maximum observed stutter ratios [30] were removed from the LRmix and Lab Retriever input files. The input files for each system are available as supplementary material.

All three of the programmes used as examples are intended for use by expert operators who may, correctly, intervene in the process at one or more points to produce a more robust answer. By testing in this way we are not examining the performance of the software operator but rather the software alone without expert guidance. Having conceded these limitations we contend that the test of functionality undertaken by these tests is still very beneficial if not vital. These tests examine some of the basic functionality of the software which, if operating incorrectly, cannot be corrected by any expert intervention.

Allele counts for a Caucasian population from Budowle et al. [31–33] were used and a θ value of 0.01 was applied.

The propositions addressed for all experiments in all three programmes were:

H_p: the sample contains the DNA of the POI and $n-1$ unknown people

H_d: the sample contains the DNA of n unknown people

where n is the number of contributors to the profile. In this experiment we suggest considering all the unknown persons to be unrelated to the POI although we do not wish to give the impression that relatedness is unimportant.

The expected likelihood ratio for all profiles was calculated in Excel using the Balding and Nichols formulae or variants developed using the sampling formula [25]. All Excel calculations are provided as supplementary material.

2.3. Experiment 1: comparison with the expected LR with no dropout or drop-in

The peak heights of the two single source Identifiler™ profiles (references 1 and 2) were adjusted so that dropout was not possible. The same two profiles were then combined to create two mixed DNA profiles, one at a 3:1 ratio (referred to as *major:minor*) and one at a 1:1 ratio (referred to as *balanced*). The peak heights of both contributors to the mixed DNA profiles were at heights where dropout was not possible. The *LR* for the major:minor profile was calculated in Excel assuming complete resolution of both contributors to the profile. The *LR* for the balanced profile was calculated using a restricted combinatorial approach using peak heights.

2.4. Experiment 2: the effect of dropout

We consider for this experiment a situation where a laboratory has set a stochastic threshold (ST). This threshold is set to be the height of a peak above which the heterozygous partner of that peak is very unlikely to have dropped out.

The minor contributor to the major: minor profile was reduced in height to an average profile peak height of 180 rfu, a level below the ST. At this level we would expect the PCR process to produce peaks with a high relative variance. We name this profile the *stochastic* profile. None of the methods use an ST to interpret the profile. STRmix™ models the variability in peak height, which is known to increase as peak height decreases. This relationship can be shown by calculating heterozygous balance variation over a range of average peak heights [24]. The semi-continuous methods incorporate a dropout probability into the *LR* calculation. Pr(D) can be assigned by logistic regression and empirical data [34,35], by using empirical data directly [36], or in the case of LRmix finding the 0.90 probability bounds for the probability of the number of alleles given the known contributors, the allele probabilities and the probability of dropout.

Lab Retriever recommends the use of a dropout probability calculator (available at http://www.scieg.org/lab_retriever.html) which optionally uses the average peak heights of the entire profile or only those of the relevant component (for example major or minor contribution) as the conditioning information, R, to determine the value of Pr($D|R$) for use in the calculation. Using the Lab Retriever calculator the two options for the selection of the conditioning information are:

1. The average height of all peaks above the analytical threshold, or
2. The average of the minor peaks.

At an analytical threshold of 50 rfu, the Pr($D|R$) assigned using option 1 was 0.000 and for option 2 was calculated at 0.016 when considering non-masked alleles corresponding to reference 2. It is likely that for both semi-continuous models the major will be identified by human judgement and reported as the single source *LR*/2 [37–42]. Within the *LR* calculation for Lab Retriever the value of Pr($D|R$) was set to 0.016 considering reference 2 (the minor) as the person of interest which appears to be the recommendation in the manual. We note however that this single value is applied to all alleles whether major or minor.

LRmix includes an optional sensitivity test function that suggests a probability of dropout that minimises the *LR* within a plausible range [27]. The final *LR* is recalculated using accordingly [43]. The sensitivity analysis was undertaken assuming the major and minor contributors to the mixture. A plot of log (*LR*) versus Pr(D) for the minor contributor (reference 2) is given in Fig. 1.

Following this method, *LR*s were calculated using Pr(D) = 0.16 for contributor 2 (minor). This procedure arises from the criterion to select a Pr(D) that minimises the *LR*. Hence for a contributor with high peak heights (e.g. the major contributor in this example), a high Pr(D) minimises the *LR*.

The probability of drop-in was set at 0.00 for all methods which is rational given the manner in which we have created the profiles.

2.5. Experiment 3: the effect of drop-in

An additional allele, allele 14, was included at 75 rfu at a heterozygote locus D8S1179 in the single source profile corresponding to reference 1 (refer Fig. 2). A probability of drop-in of 0.01 was used for the semi-continuous methods and Pr(D) was set to 0.00. Drop-in was modelled using an exponential model within STRmix™ [11].

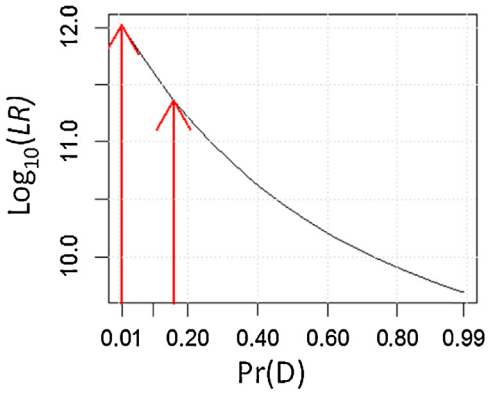


Fig. 1. LRmix sensitivity analysis plotting log (*LR*) versus Pr(*D*) for the stochastic mixture assuming reference 2 is known under H_p.

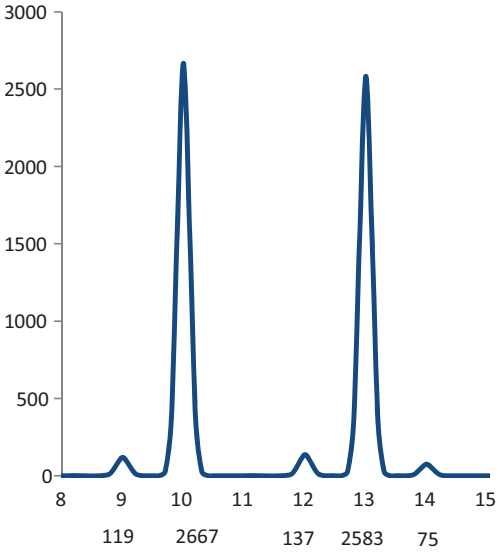


Fig. 2. Heterozygote locus from a single source profile with one drop-in 14 allele. Heights are given below the allele in rfu.

2.6. Experiment 4: reproducibility

Unlike Lab Retriever and LRmix, STRmix™ will produce a different *LR* from each replicate analysis of identical input data. STRmix™ employs a Markov chain Monte Carlo (MCMC) method to build the genotype weight at each locus. Where an unambiguous profile cannot be inferred from a profile, variation is introduced. The reproducibility of the STRmix™ software is examined by analysing the stochastic profile from Experiment 2 ten times and comparing the *LR* for both contributors across all replicates. All profiles were interpreted with 40,000 MCMC accepts after a period of 10,000 burn-in accepts.

3. Results

The *LR*s reported here are strictly conditioned on simulated allele-sets and two alternative propositions. However, in real casework, within the 'exploratory approach' described in Gill and Haned [14] there is a process of evaluative pre-assessment which includes analysis of the peak heights and the option to examine multiple sets of propositions. Therefore, the *LR*s listed in Tables 1–3 do not necessarily reflect the *LR*s that would actually be reported in a real case.

The results of Experiment 1 appear in Table 1. The *LR* for reference 2 (minor) to the major:minor mixed profile is 2.44×10^{19}. This assumes complete resolution of the minor which is unlikely therefore we expect a result that is less than 2.44×10^{19}.

The results of Experiments 1 and 2 appear in Tables 1 and 2, respectively.

The *LR* was not calculated for the minor contributor to major:minor profile where dropout was possible (Experiment 2). The *LR* for the major contributor (reference 1) is similar to the *LR*s from Experiment 1. The *LR* for the minor contributor (reference 2) has decreased for all programmes tested. This is the expected result. An *LR* of 2.44×10^{19} assumes that the minor has been completely resolved which is not likely. The *LR* for the minor where dropout is possible (stochastic) would be significantly less than 2.44×10^{19}, which is indicated by $<<$.

The effect of drop-in at one locus on the *LR* at that locus for all software is demonstrated in Table 3.

The results of the replicate interpretations using STRmix™ are in Fig. 3. A summary of the values are provided in Table 4. As expected the *LR* for the major contributor is consistent between the replicates. The *LR* for the minor contributor is more variable between replicate interpretations.

4. Discussion

4.1. Experiment 1

Profiles and allele counts are available as supplementary material so that inspection of these results can be undertaken.

For a single source profile we can reasonably expect exact concordance between the software and the expected result. STRmix™ and LRmix produce the expected result. Lab Retriever produces a result very close to the expected result. STRmix™ and LRmix implement the Balding and Nichols' equations also known as recommendation 4.2 of NRC II [25,26]. Lab Retriever implements an approximation present in the Balding and Buckleton paper [13] (see Appendix 1).

Table 1
The results for the comparisons where dropout and drop-in is not possible (Experiment 1).

Software or method	Single source		Major:minor		Balanced	
	Reference 1	Reference 2	Reference 1	Reference 2	Reference 1	Reference 2
LRmix	6.06×10^{18}	4.31×10^{19}	3.03×10^{18}[a]	1.83×10^{16}[b]	1.02×10^{9}	9.14×10^{9}
Lab Retriever	6.01×10^{18}	4.36×10^{19}	3.00×10^{18}[a]	1.76×10^{16}[b]	9.90×10^{8}	8.94×10^{9}
STRmix™	6.06×10^{18}	4.31×10^{19}	3.39×10^{18}	2.03×10^{19}	6.54×10^{9}	4.45×10^{10}
Excel	6.06×10^{18}	4.31×10^{19}	3.40×10^{18}	$<2.44 \times 10^{19}$	6.59×10^{9}	4.48×10^{10}

[a] It is likely that for the semi-continuous models the major will be identified by human judgement and reported as the single source *LR*/2. However we do not know what normal practice would be with respect to the conditioning alleles for the Balding and Nichol's calculations. If all the alleles of the POI and the profile were used then these values would both be 3.40×10^{18}.

[b] We have applied a manual adjustment described in Appendix 2 that works for either LRmix or Lab Retriever and recovers much of the information present in heights.

Table 2
The results for the comparisons where dropout is possible (Experiment 2). We reproduce part of Table 1 for ease of comparison.

Software or method	Stochastic			
	Dropout not possible	Dropout possible	Dropout not possible	Dropout possible
	Reference 1		Reference 2	
LRmix	3.03×10^{18}		1.83×10^{16}	2.29×10^{11}[a]
Lab Retriever	3.00×10^{18}		1.76×10^{16}	1.01×10^{12}[a]
STRmix™	3.39×10^{18}	3.08×10^{18}	2.03×10^{19}	4.88×10^{15}
Excel	3.40×10^{18}	3.40×10^{18}	$<2.44 \times 10^{19}$	$<<2.44 \times 10^{19}$

[a] This *LR* is obtained by using the stutter artefact as a masking allele. Ambiguous alleles which are in stutter positions of the major (reference 1) may be assigned to an additional (masking) profile under both H_p and H_d. See Appendix 3 for details.

Table 3
The *LR* at D8S1179 with drop-in of one allele.

Software or method	Without drop-in	With drop-in	Excel with drop-in
LRmix	13.41	4.65	4.66
Lab Retriever	13.28	4.42	4.66
STRmix™	13.41	13.41	13.41

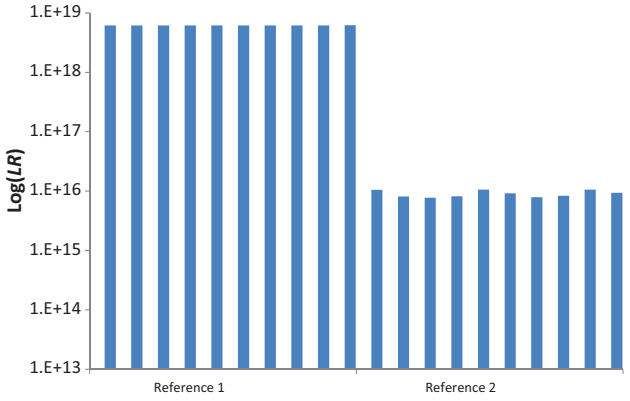


Fig. 3. Plot of ten replicate analyses using STRmix™ for the major:minor mixture assuming reference 1 and then 2 as the POI under H_p.

Table 4
Summary of *LR* for ten replicate STRmix™ analyses.

	Major	Minor
Mean	6.2×10^{18}	9.04×10^{15}
Standard deviation	1.5×10^{16}	1.16×10^{15}
Coefficient of variation (SD/mean)	0.24%	12.85%

For the major:minor mixture STRmix™ gets close to the expected answer as do the other two if the major is extracted manually.

For the balanced profile all three programmes produce close to the expected result and to each other. For such profiles the information content of peak height has a lesser effect than for a major:minor mixture and hence the two semi-continuous programmes are not suffering from information loss with respect to STRmix™.

4.2. Experiment 2

In Experiment 2 the peak heights for the minor contributor have been reduced so that the alleles fall below the nominated ST. The peak heights of the major profile are still well above the ST and should be unaffected. In fact they should be close to the single source *LR* for reference 1, divided by two [37]. STRmix™ obtains an *LR* very close to this expected result (Table 3). The difference between the result with and without dropout is probably the Monte Carlo variability inherent in STRmix™. The other two programmes also obtain a value close to or equal to this when the major is extracted manually.

For reference 2, the minor, we expect a small reduction in the *LR* compared with the profile where dropout is not possible. This is observed for all software.

4.3. Experiment 3

Since the profile is described as single source and there are three peaks, all three methods should treat one peak as drop-in. STRmix™, which uses peak height information, will pair the 10 and 13 alleles, which appear at heights above 2500 rfu, and treat the 14 as a drop-in. Lab Retriever and LRmix however do not use peak heights and therefore will interpret the locus three different ways, each with the same probability: a 10,13 heterozygote with the 14 allele as drop-in, 10,14 with the 13 allele as drop-in, and 13,14 with the 10 allele as drop-in.

STRmix™ results in an *LR* for this locus that is the same as without drop-in since the exact same drop-in event is required under H_p and H_d and therefore cancels in the calculation. The two semi-continuous methods are expected to give an *LR* of 4.66 (refer to the calculations in the supplementary material) and gave results close to this. Again this difference between STRmix™ and the semi-continuous systems is a reflection of the use, or non-use, of peak heights and not an error in the programmes.

4.4. Experiment 4

The run to run variation inherent in the Monte Carlo process at the heart of STRmix™ means that for unambiguous genotype combinations a different result is obtained for replicate analyses. The magnitude of the variation depends mainly on two factors; the size of the weights for genotype sets in H_p and the length the MCMC analysis is run [44,45].

5. Conclusion

An understanding of the models within each of the programmes and their limitations is required in order to validate interpretation software. There are a limited number of examples where the expected answer can be obtained relatively easily using Excel, or similar. This process can be facilitated by reducing complicating factors such as setting θ (or Fst) to zero, which should return the product rule, and/or setting Pr(D) and Pr(C) to zero. Note that currently θ is hardcoded in Lab Retriever at 0.01. Additional functionality checks include evaluating the effect of changing θ values within the same population and changing allele frequency tables, where possible.

Gaining an understanding of the behaviour of the software under certain conditions is central to the developmental validation process prior to use in casework. Gaining an understanding of this work is an important part of the internal validation and training process prior to implementation of software. This includes calibration based on ground-truth cases where the contributors are known and case hardening to test how a programme performs in the real world. As an example, in a comprehensive study using STRmix™, Taylor [44] examines the effect of profile complexity, replicate amplifications, assumed contributors, incorrect and irrelevant information has on the *LR*. In addition, tests of the method when H_d is true are described by Gill and Haned [14] (non-contributor performance tests) and Taylor et al. [11,46].

We have proposed a number of tests that can be undertaken as part of an internal validation of profile interpretation software. We would suggest that an essential feature of validation is the ability to specify exactly what the software is doing at least with regard to routine matters such as the population genetic model and the allele probabilities. We have made the distinction from the requirements of developmental validation. The QAS requires the peer-reviewed publication of the underlying scientific principles of a technology. We advocate that for forensic DNA profile interpretation software these publications should include at least the population genetic models used within the programme, justification for the models and demonstration of the performance under controlled conditions.

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Appendix 1. Lab Retriever is an implementation of an R-programme initially produced by David Balding

The programme references Balding and Buckleton for the mathematics [13]. Using the terms CSP for crime stain profile and *s* for suspect, that paper states: "To allow approximately for the effects of remote shared ancestry between *s* and other possible sources of the CSP, we replace the allele proportion *P*, adjusted for sampling error as described above, with $(\theta + (1-\theta)p)/1+\theta$ for each allele when *s* is heterozygote, and with $(2\theta + (1-\theta)p)/1+\theta$ in the homozygote case. For alleles not in the profile of *s*, the allele proportion *P* is replaced with $((1-\theta)p)/1+\theta$."

These equations are approximately but not exactly those given in Balding and Nichols [25]. As a consequence the formulae used within Lab Retriever differ and are replicated here.

CSP	POI	Balding and Nichols' Formula [25]	Lab Retriever following Balding and Buckleton [13]
aa	aa	$\frac{[3\theta+(1-\theta)p_a][2\theta+(1-\theta)p_a]}{(1+\theta)(1+2\theta)}$	$\frac{[2\theta+(1-\theta)p_a][2\theta+(1-\theta)p_a]}{(1+\theta)(1+\theta)}$
ab	ab	$\frac{2[\theta+(1-\theta)p_a][\theta+(1-\theta)p_b]}{(1+\theta)(1+2\theta)}$	$\frac{2[\theta+(1-\theta)p_a][\theta+(1-\theta)p_b]}{(1+\theta)(1+\theta)}$

Appendix 2. This workaround is attributable to Carlos Bieza Richer

If we consider a mixture where there is a clear major profile and one is interested in interpreting the minor then one can add a known contributor that is the major in the numerator and the denominator. This will allow interpretation of the minor profile except for the factor of 2 (or $N!$)) [37–42].

Appendix 3. This adjustment follows that suggested in Balding and Buckleton 2009 and has been implemented in Lab Retriever and may be applied in LRmix

If we consider some alleles which may be stutter or may be allelic it is useful to treat them as ambiguous. This may be achieved by creating an artificial profile containing only those alleles that are to be considered ambiguous. There may be zero to many of these per locus. This profile is added as an assumed contributor under both H_p and H_d.

Note that within LRmix assumed reference profiles must be complete otherwise the locus *LR* is 0. When there are 0 or 1 alleles designated the additional alleles may be 'hidden under' the major profile by simply assigning the ambiguous alleles as the same as the major alleles.

Appendix D. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.fsigen.2014.09.019.

References

[1] C. Vecchiotti, S. Conti, The Connti-Vecchiotti Report, 2011 Available from: http://knoxdnareport.wordpress.com/, (cited 5 May 2014).
[2] A. Haelser, Issues in gathering, interpreting and delivering DNA evidence, in: Expert Evidence Conference, Canberra, 2011.
[3] L. Geddes, Fallible DNA evidence can mean prison or freedom, New Sci. 207 (2773) (2010) 8–11.
[4] J. Butler, M. Kline, NIST Mixture Interpretation Interlaboratory Study 2005 (MIX05), in: Promega's Sixteenth International Symposium on Human Identification, Grapevine, TX, 2005.
[5] Scientific Working Group on DNA Analysis Methods (SWGDAM), SWGDAM Interpretation Guidelines for Autosomal STR Typing by Forensic DNA Testing Laboratories, 2010 Available from: http://www.fbi.gov/hq/lab/html/codis_swgdam.pdf.
[6] N. Morling, et al., Interpretation of DNA mixtures – European consensus on principles, Forensic Sci. Int. Genet. 1 (3–4) (2007) 291–292.
[7] P. Gill, et al., DNA commission of the International Society of Forensic Genetics: recommendations on the interpretation of mixtures, Forensic Sci. Int. 160 (2006) 90–101.
[8] P. Gill, et al., DNA commission of the International Society of Forensic Genetics: recommendations on forensic analysis using Y-chromosome STRs, Int. J. Legal Med. 114 (2001) 305–309.
[9] P. Gill, et al., DNA commission of the International Society of Forensic Genetics: recommendations on the evaluation of STR typing results that may include drop-out and/or drop-in using probabilistic methods, Forensic Sci. Int. Genet. 6 (6) (2012) 679–688.
[10] A. Carracedo, et al., Focus issue – analysis and biostatistical interpretation of complex and low template DNA samples, Forensic Sci. Int. Genet. 6 (6) (2012) 677–678.
[11] D. Taylor, J.-A. Bright, J. Buckleton, The interpretation of single source and mixed DNA profiles, Forensic Sci. Int. Genet. 7 (5) (2013) 516–528.
[12] M.W. Perlin, et al., Validating TrueAllele® DNA mixture interpretation, J. Forensic Sci. 56 (6) (2011) 1430–1447.
[13] D.J. Balding, J. Buckleton, Interpreting low template DNA profiles, Forensic Sci. Int. Genet. 4 (1) (2009) 1–10.
[14] P. Gill, H. Haned, A new methodological framework to interpret complex DNA profiles using likelihood ratios, Forensic Sci. Int. Genet. 7 (2) (2013) 251–263.
[15] H. Kelly, et al., A comparison of statistical models for the analysis of complex forensic DNA profiles, Sci. Justice 54 (2014) 66–70.
[16] C.D. Steele, D.J. Balding, Statistical evaluation of forensic DNA profile evidence, Annu. Rev. Stat. Appl. 1 (2014) 20.1–20.24.
[17] Federal Bureau of Investigation, Quality Assurance Standards for Forensic DNA Testing Laboratories, 2011, September 1 Available from: http://www.fbi.gov/about-us/lab/codis/qas-standards-for-forensic-dna-testing-laboratories-effective-9-1-2011, (cited 30 March 2014).

[18] Scientific Working Group on DNA Analysis Methods (SWGDAM), Validation Guidelines for DNA Analysis Methods, 2012 Available from: http://swgdam.org/SWGDAM_Validation_Guidelines_APPROVED_Dec_2012.pdf, (cited 30 March 2014).
[19] H. Haned, Forensim an open-source initiative for the evaluation of statistical methods in forensic genetics, Forensic Sci. Int. Genet. 5 (4) (2011) 265–268.
[20] K. Lohmueller, N. Rudin, Calculating the weight of evidence in low-template forensic DNA casework, J. Forensic Sci. 58 (1) (2013) 234–259.
[21] P. Gill, J. Curran, K. Elliot, A graphical simulation model of the entire DNA process associated with the analysis of short tandem repeat loci, Nucl. Acids Res. 33 (2) (2005) 632–643.
[22] J.-A. Bright, et al., Degradation of forensic DNA profiles, Aust. J. Forensic Sci. 45 (4) (2013) 445–449.
[23] C. Brookes, et al., Characterising stutter in forensic STR multiplexes, Forensic Sci. Int. Genet. 6 (1) (2012) 58–63.
[24] J.-A. Bright, et al., Developing allelic and stutter peak height models for a continuous method of DNA interpretation, Forensic Sci. Int. Genet. 7 (2) (2013) 296–304.
[25] D.J. Balding, R.A. Nichols, DNA profile match probability calculation: how to allow for population stratification, relatedness, database selection and single bands, Forensic Sci. Int. 64 (1994) 125–140.
[26] NRC II, National Research Council Committee on, DNA. Forensic Science, The Evaluation of Forensic, DNA Evidence, National Academy Press, Washington, DC, 1996.
[27] P. Gill, A. Kirkham, J. Curran, LoComatioN: a software tool for the analysis of low copy number DNA profiles, Forensic Sci. Int. 166 (2–3) (2007) 128–138.
[28] P. Gill, et al., National recommendations of the technical UK DNA working group on mixture interpretation for the NDNAD and for court going purposes, Forensic Sci. Int. Genet. 2 (1) (2008) 76–82.
[29] P. Gill, J. Buckleton, A universal strategy to interpret DNA profiles that does not require a definition of low-copy-number, Forensic Sci. Int. Genet. 4 (4) (2010) 221–227.
[30] Applied Biosystems, AmpFlSTR Identifiler Amplification Kit User Guide, Foster City, CA, Applied Biosystems, 2009.
[31] B. Budowle, et al., Population data on the thirteen CODIS core short tandem repeat loci in African Americans, US, Caucasians, Hispanics, Bahamanians, Jamaicans and Trinidadians, J. Forensic Sci. 44 (1999) 1277–1286.
[32] B. Budowle, et al., Population data on the STR Loci D2S1338 and D19S433, Forensic Sci. Commun. 3 (3) (2001), Available at http://www.fbi.gov/about-us/lab/forensic-science-communications/fsc/july2001/index.htm/budowle2.htm.
[33] B. Budowle, T.R. Moretti, Genotype profiles for six population groups at the 13 CODIS short tandem repeat core loci and other PCR-based loci, 1999 Available from: http://www.fbi.gov/programs/lab/fsc/backissu/july1999, (cited July 1999).
[34] T. Tvedebrink, et al., Estimating the probability of allelic drop-out of STR alleles in forensic genetics, Forensic Sci. Int. Genet. 3 (4) (2009) 222–226.
[35] J. Buckleton, et al., Utilising allelic dropout probabilities estimated by logistic regression in casework, Forensic Sci. Int. Genet. 9 (1) (2014) 9–11.
[36] A.A. Mitchell, et al., Likelihood ratio statistics for DNA mixtures allowing for drop-out and drop-in, Forensic Sci. Int. Genet. Suppl. Series 3 (1) (2011) e240–e241.
[37] C.M. Triggs, J. Buckleton, The two trace transfer problem revisited, Sci. Justice 43 (3) (2003) 127–134.
[38] D.A. Stoney, Relaxation of the assumption of relevance and an application to one-trace and two-trace problems, J. Forensic Sci. Soc. 34 (1994) 17–21.
[39] S. Gittelson, et al., Bayesian networks and the value of the evidence for the forensic two-trace transfer problem, J. Forensic Sci. 57 (5) (2012) 1199–1216.
[40] S. Gittelson, et al., Modeling the forensic two-trace problem with Bayesian networks, Artif. Intell. Law 21 (2) (2013) 221–252.
[41] I.W. Evett, On meaningful questions: a two-trace transfer problem, J. Forensic Sci. Soc. 27 (1987) 375–381.
[42] R. Meester, M. Sjerps, The evidential value in the DNA database search controversy and the two-stain problem, Biometrics 59 (3) (2003) 727–732.
[43] L. Prieto, et al., Euroforgen-NoE collaborative exercise on LRmix to demonstrate standardization of the interpretation of complex DNA profiles, Forensic Sci. Int. Genet. 9 (2014) 47–54.
[44] D. Taylor, Using continuous DNA interpretation methods to revisit likelihood ratio behaviour, Forensic Sci. Int. Genet. 11 (2014) 144–153.
[45] D. Taylor, et al., An illustration of the effect of various sources of uncertainty on DNA likelihood ratio calculations, Forensic Sci. Int. Genet. 11 (2014) 56–63.
[46] J.-A. Bright, et al., Searching mixed DNA profiles directly against profile databases, Forensic Sci. Int. Genet. 9 (2014) 102–110.